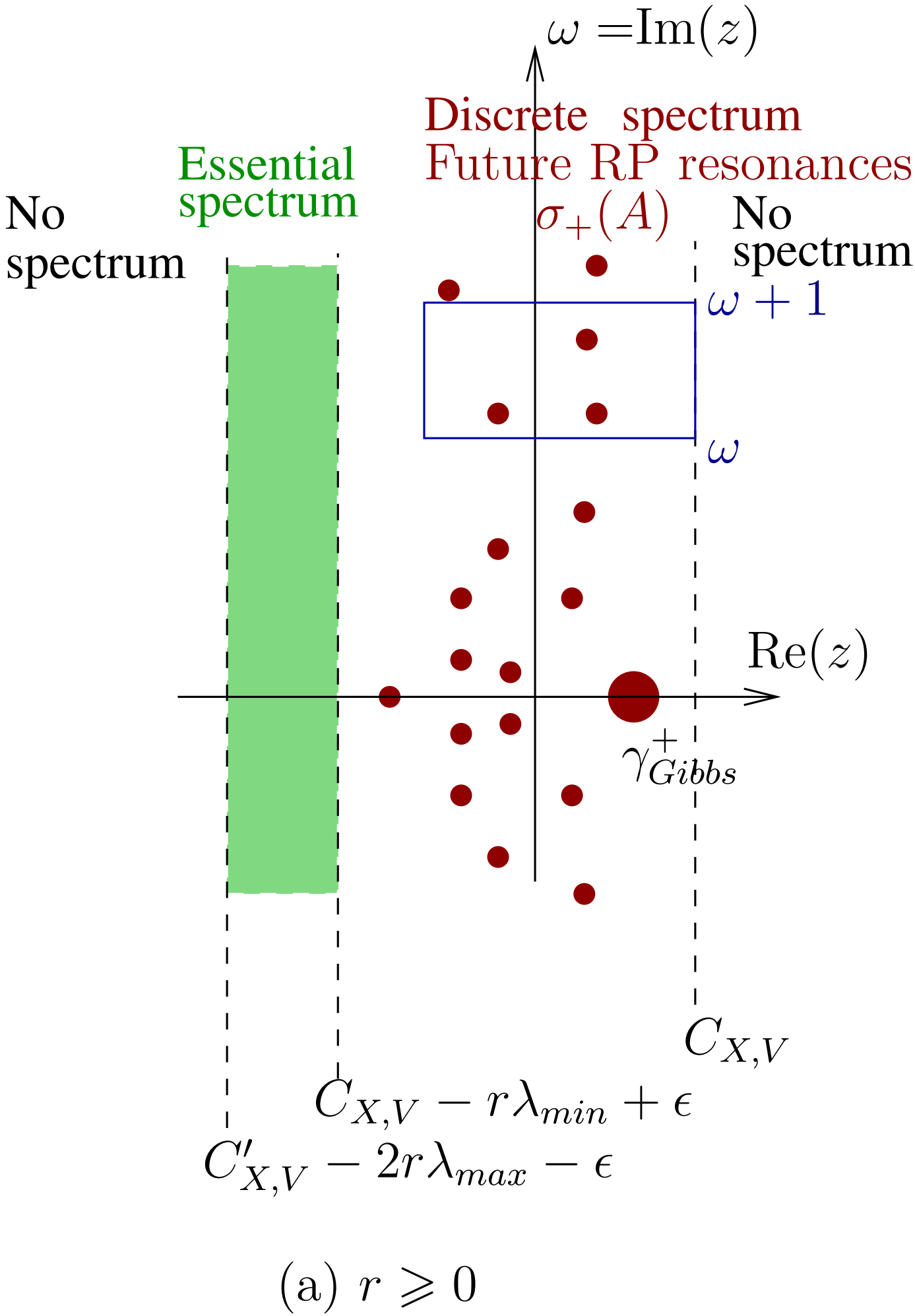


(a) $r \geqslant 0$